Topics in Algebra
Banach Tarski Paradox

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Abstract

This report gives a self-contained proof of the Banach–Tarski paradox. We outline the foundational context, develop the required group-theoretic machinery, construct a paradoxical decomposition of the sphere, and extend it to a solid ball using equidecomposability. The proof illustrates how free group actions and the Axiom of Choice combine to produce the classical paradox.

Keywords: Banach–Tarski Paradox, Axiom of Choice, Measure Theory, Paradoxical Decomposition

1 Introduction

The problem of assigning a meaningful notion of "size" to arbitrary subsets of $\mathbb{R}^n$ lies at the heart of measure theory and modern analysis. Classical intuition suggests that such a notion should extend the familiar geometric concepts of length, area, and volume, and should respect the underlying symmetries of Euclidean space. One is therefore naturally led to seek a measure μ defined on all subsets of $\mathbb{R}^n$ that satisfies the following basic requirements:

1. *countable additivity* on disjoint families;
2. *invariance* under rigid motions;
3. *normalization* on standard sets, such as $\mu([0,1]^n) = 1$.

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A classical argument shows that these requirements are incompatible. By using the axiom of choice to select one representative from each orbit under rational translation, one can partition the unit interval in a way that forces any countably additive, translation-invariant measure to assign contradictory values. Thus no such measure can exist on all subsets of $\mathbb{R}$.

In higher dimensions an even more striking phenomenon occurs. In 1924 Banach and Tarski [1] showed that in $\mathbb{R}^n$ for $n \geq 3$, one may dispense with countable additivity entirely and still encounter impossibility. They proved that any two bounded open sets in $\mathbb{R}^3$ are *equidecomposable*: each can be partitioned into finitely many pieces that can be rearranged by rigid motions to form the other. In particular, a solid ball can be divided into finitely many disjoint, highly nonmeasurable pieces, which may then be reassembled into two balls congruent to the original. A sharp result due to Robinson [2] shows that such a doubling of the ball can be accomplished with *exactly* five, and fewer than five pieces will not suffice.

At the conceptual core of the Banach–Tarski paradox is the idea of a *paradoxical decomposition*: a set that can be partitioned into finitely many pieces which, after applying suitable isometries, form two disjoint copies of the original set. Tarski later demonstrated that such decompositions occur precisely for group actions containing a nonabelian free subgroup. This insight places group theory—particularly the behavior of free groups and their actions—at the center of the phenomenon. The classical proof constructs a paradoxical decomposition of the free group F_2, embeds F_2 into $\mathrm{SO}(3)$, and uses this embedding to transfer the paradox to the unit sphere S^2. A radial extension then propagates the decomposition to the entire unit ball.

The purpose of this report is to give a streamlined, self-contained exposition of this strategy. In Section 2 we review the relevant concepts from group actions, equidecomposability, and paradoxical decompositions. Section 3 develops the proof: we obtain a paradoxical decomposition of the free group F_2, embed it into $\mathrm{SO}(3)$, transfer the resulting decomposition to S^2, and extend it to the solid unit ball via radial invariance. The presentation emphasizes the interplay between group-theoretic structure and the axiom of choice, showing how these ingredients combine to yield one of the most remarkable results in modern set theory and geometric analysis.

2 Preliminaries

In this section we introduce the basic notions required for the Banach–Tarski paradox. Throughout, G denotes a group and X a set. Our goal is to understand how group actions allow us to "move pieces around," and how such motions lead to equidecompositions and paradoxical decompositions.

Definition 2.1 (Group Action). *A* (left) action *of G on X is a function*

$$G \times X \longrightarrow X, \qquad (g, x) \mapsto g \cdot x,$$

such that for all $g, h \in G$ and all $x \in X$,

1. *$e \cdot x = x$, where e is the identity element of G;*

2. *$(gh) \cdot x = g \cdot (h \cdot x)$.*

In this situation we say that G acts *on X.*

Remark 2.1. *A* right *action is defined similarly using a function $X \times G \to X$ and the rule $(x \cdot g) \cdot h = x \cdot (gh)$.*

Definition 2.2 (Equidecomposability)**.** *Let G act on a set X. Two subsets $A, B \subseteq X$ are G-*equidecomposable *if there exist pairwise disjoint subsets $A_1, \dots, A_n \subseteq A$ and elements $g_1, \dots, g_n \in G$ such that*

$$B = \bigcup_{i=1}^{n} g_i \cdot A_i.$$

In this case we write $A \sim_G B$ and say that A can be partitioned into finitely many pieces and moved by elements of G to form B.

Definition 2.3 (Paradoxical Decomposition)**.** *Let G act on X. A subset $A \subseteq X$ admits a* paradoxical decomposition *if there exist pairwise disjoint subsets*

$$A_1, \dots, A_m, \qquad B_1, \dots, B_n \subseteq A$$

and elements

$$g_1, \dots, g_m, \qquad h_1, \dots, h_n \in G$$

such that

$$A = \bigcup_{i=1}^{m} g_i \cdot A_i \qquad \text{and} \qquad A = \bigcup_{j=1}^{n} h_j \cdot B_j.$$

That is, A can be partitioned into finitely many pieces which, after applying elements of G, form two disjoint copies of A.

Lemma 2.1. *If $A \sim_G B$ and A admits a paradoxical decomposition under G, then B also admits a paradoxical decomposition under G.*

Proof. Suppose

$$A = \bigcup_{i=1}^{m} g_i A_i = \bigcup_{j=1}^{n} h_j B_j$$

is paradoxical, and that $A \sim_G B$ via a decomposition

$$B = \bigcup_{k=1}^{r} t_k C_k, \qquad C_k \subseteq A.$$

Replacing each A_i and B_j by its intersections with the C_k (and then applying the corresponding t_k) yields a paradoxical decomposition of B. ■

Remark 2.2. *Removing a countable subset does not affect equidecomposability or paradoxicality. If $D \subset X$ is countable and G acts on X, then X is G-equidecomposable with $X \setminus D$. This follows from the fact that a countable set can be "absorbed" into a single piece of any finite equidecomposition by sending its points into an infinite orbit.*

A paradoxical decomposition is the formal mechanism behind the Banach–Tarski paradox. In Theorem 3.2 we will show that the closed unit ball $B = \{x \in \mathbb{R}^3 : \|x\| \leq 1\}$ admits a paradoxical decomposition under the action of the group of rotations. In particular, B is equidecomposable with two disjoint copies of itself.

3 The Banach–Tarski Paradox

In this section we give a detailed proof of the Banach–Tarski paradox following the classical strategy:

1. construct a paradoxical decomposition of F_2; And realize this free group as a subgroup of the rotation group SO(3);
2. transfer the paradoxical decomposition from F_2 to the unit sphere S^2;
3. extend the paradoxical decomposition of S^2 to the solid unit ball in $\mathbb{R}^3$.

Throughout, G-equidecomposability and paradoxical decomposition are meant in the sense introduced in Section 2.

3.1 A paradoxical decomposition of the free group F_2

Let F_2 be the free group on generators a and b, equipped with its standard description in terms of reduced words.

Definition 3.1. *A word $w = x_1 x_2 \cdots x_n$ in the alphabet $\{a, a^{-1}, b, b^{-1}\}$ is* reduced *if no letter is adjacent to its inverse. The free group F_2 is the set of reduced words together with the empty word e, with multiplication given by concatenation followed by cancellation of all occurrences of xx^{-1} and $x^{-1}x$.*

Every nontrivial element of F_2 begins with exactly one of the letters a, a^{-1}, b, or b^{-1}. Define the corresponding subsets:

$$\begin{aligned} S(a) &= \{\, w \neq e : w \text{ begins with } a \,\}, \\ S(a^{-1}) &= \{\, w \neq e : w \text{ begins with } a^{-1} \,\}, \\ S(b) &= \{\, w \neq e : w \text{ begins with } b \,\}, \\ S(b^{-1}) &= \{\, w \neq e : w \text{ begins with } b^{-1} \,\}. \end{aligned}$$

These four sets are disjoint and partition all nontrivial words, giving

$$F_2 = \{e\} \cup S(a) \cup S(a^{-1}) \cup S(b) \cup S(b^{-1}). \tag{1}$$

For any $w \in S(a^{-1})$ we may write $w = a^{-1}u$, so $aw = aa^{-1}u \to u$ after reduction. Conversely, every reduced word not beginning with a arises uniquely in this way. Applying the same reasoning for b, we obtain two disjoint decompositions:

$$F_2 = S(a) \cup aS(a^{-1}), \qquad \text{and} \qquad F_2 = S(b) \cup bS(b^{-1}).$$

Each of these equalities expresses F_2 as a finite union of disjoint pieces that are moved by a single group element (namely a or b) to reassemble the whole group. Taken together, they give two different reassemblies of F_2 from finitely many translated pieces.

Corollary 3.1. *The free group F_2, acting on itself by left translation, admits a paradoxical decomposition. In particular, F_2 is F_2-paradoxical.*

Theorem 3.1. *There exist rotations $A, B \in \mathrm{SO}(3)$ such that the subgroup $H = \langle A, B\rangle$ is isomorphic to F_2.*

Sketch. Choose two orthogonal axes and a rotation angle $\theta = \arccos(1/3)$. Let A and B be rotations by θ about these axes. A classical coordinate argument shows that no nontrivial reduced word in $A^{\pm 1}, B^{\pm 1}$ fixes the point $(1, 0, 0)$, hence no reduced word represents the identity rotation. Thus $\langle A, B\rangle$ is a free subgroup on two generators. ■

We henceforth identify F_2 with its copy $H = \langle A, B\rangle \subset \mathrm{SO}(3)$.

3.2 A paradoxical decomposition of the sphere

Let S^2 be the unit sphere. The action of H on S^2 is not free everywhere: rotations have two fixed points each.

Lemma 3.1. *Let $D \subset S^2$ be the set of points fixed by some nontrivial element of H. Then D is countable.*

Proof. Every nontrivial rotation fixes exactly two antipodal points, and H is countable. Thus D is a countable union of two-point sets. ■

Let $S_0^2 := S^2 \setminus D$. The action of H on S_0^2 is free. By the axiom of choice, we may select a set $M \subset S_0^2$ containing exactly one representative from each H-orbit. The map

$$H \times M \to S_0^2, \qquad (h, m) \mapsto h \cdot m$$

is bijective. Let

$$H = \bigcup_{i=1}^{k} E_i$$

be the finite partition of H. For each i define

$$E_i \cdot M := \{h \cdot m : h \in E_i,\ m \in M\}.$$

Then sets $E_i \cdot M$ form a partition of S_0^2.

Proposition 3.1. *S_0^2 admits a paradoxical decomposition under the action of H.*

Proof. Apply the paradoxical decomposition of H and transfer each piece E_i to $E_i \cdot M$. The relations are preserved because the action of H on S_0^2 is free. ■

Proposition 3.2. *S_0^2 admits a paradoxical decomposition under the action of H.*

Proof. Transfer the paradoxical decomposition of H to S_0^2 using the bijection $H \times M \to S_0^2$. We obtain disjoint families

$$A_i \cdot M, \qquad B_j \cdot M$$

whose translates satisfy

$$S_0^2 = \bigcup_{i=1}^{m} g_i(A_i \cdot M) \qquad \text{and} \qquad S_0^2 = \bigcup_{j=1}^{n} h_j(B_j \cdot M),$$

which is a paradoxical decomposition of S_0^2. ■

To finish, we restore the omitted pole points; being countable, they do not affect equidecomposability.

Lemma 3.2. *S^2 and S_0^2 are H-equidecomposable.*

Proof. D is countable, so by Remark 2.2, removing D does not affect equidecomposability. Thus

$$S^2 \sim_H S^2 \setminus D = S_0^2.$$

■

Corollary 3.2. *The unit sphere S^2 admits a paradoxical decomposition under rotations.*

3.3 From the sphere to the ball

We now extend the paradoxical decomposition of the sphere to the closed unit ball $B = \{x \in \mathbb{R}^3 : \|x\| \leq 1\}$.

Lemma 3.3. *If $S^2 = \bigcup A_i$ is paradoxical under rotations, then the sets*

$$\widetilde{A}_i := \{rx : x \in A_i,\ 0 < r \leq 1\}$$

form a paradoxical decomposition of $B \setminus \{0\}$.

Proof. Every point of $B \setminus \{0\}$ has a unique radial representation $x = ru$ with $u \in S^2$ and $0 < r \leq 1$. Rotations preserve radial lines: $R(rx) = r\,R(x)$. Hence any paradoxical decomposition of S^2 lifts along radial segments to a paradoxical decomposition of $B \setminus \{0\}$. ■

Finally we reincorporate the center point.

Lemma 3.4. *B and $B \setminus \{0\}$ are equidecomposable.*

Proof. The removed set $\{0\}$ is countable, so by Remark 2.2 we have $B \sim B \setminus \{0\}$. ■

Theorem 3.2 (Banach–Tarski Paradox)**.** *The closed unit ball $B \subset \mathbb{R}^3$ admits a paradoxical decomposition under rotations. In particular, B is equidecomposable with two disjoint copies of itself.*

Proof. By Corollary 3.2, the sphere S^2 is paradoxical. Radial extension gives a paradoxical decomposition of $B \setminus \{0\}$. By Lemma 3.4, $B \sim B \setminus \{0\}$. Since paradoxicality is preserved under equidecomposability (Lemma 2.1), the ball B itself is paradoxical. ■

References

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[2] Raphael M. Robinson. On the decomposition of spheres. *Fundamenta Mathematicae*, 34:246–260, 1947. `doi:10.4064/fm-34-1-246-260`.